Numerical Analysis I

Quiz 2A — Answer Key

Q2-01-A

We consider

$$f(x) = \sqrt{x}, \qquad x_0 = 1.$$

1. Taylor polynomial of degree 3

The derivatives are

$$\begin{aligned} f(x) &= x^{1/2}, \\ f'(x) &= \frac{1}{2}x^{-1/2}, \\ f''(x) &= -\frac{1}{4}x^{-3/2}, \\ f'''(x) &= \frac{3}{8}x^{-5/2}. \end{aligned}$$

At $x_0 = 1$,

$$f(1) = 1, \quad f'(1) = \frac{1}{2}, \quad f''(1) = -\frac{1}{4}, \quad f'''(1) = \frac{3}{8}.$$

Thus,

$$\boxed{P_3(x) = 1 + \frac{1}{2}(x-1) - \frac{1}{8}(x-1)^2 + \frac{1}{16}(x-1)^3.}$$

2. Error formula and bounds

Taylor's theorem with Lagrange remainder gives

$$f(x) - P_3(x) = \frac{f^{(4)}(\xi)}{4!}(x-1)^4,$$

where ξ lies between 1 and x.

Since

$$f^{(4)}(x) = -\frac{15}{16}x^{-7/2},$$

we obtain

$$\boxed{f(x) - P_3(x) = -\frac{5}{128}\xi^{-7/2}(x-1)^4.}$$

At $x = 0.9$:

$$P_3(0.9) = 1 - \frac{0.1}{2} - \frac{0.1^2}{8} - \frac{0.1^3}{16} = 0.9486875.$$

Since $0.9 \le \xi \le 1$,

$$\begin{aligned} |f(0.9) - P_3(0.9)| &\le \frac{5}{128}(0.9)^{-7/2}(0.1)^4 \\ &\approx \boxed{5.65 \times 10^{-6}}. \end{aligned}$$

At $x = 1.1$:

$$P_3(1.1) = 1 + \frac{0.1}{2} - \frac{0.1^2}{8} + \frac{0.1^3}{16} = 1.0488125.$$

Since $1 \leq \xi \leq 1.1$,

$$\begin{aligned} |f(1.1) - P_3(1.1)| &\leq \frac{5}{128}(1)^{-7/2}(0.1)^4 \\ &= \boxed{3.90625 \times 10^{-6}}. \end{aligned}$$